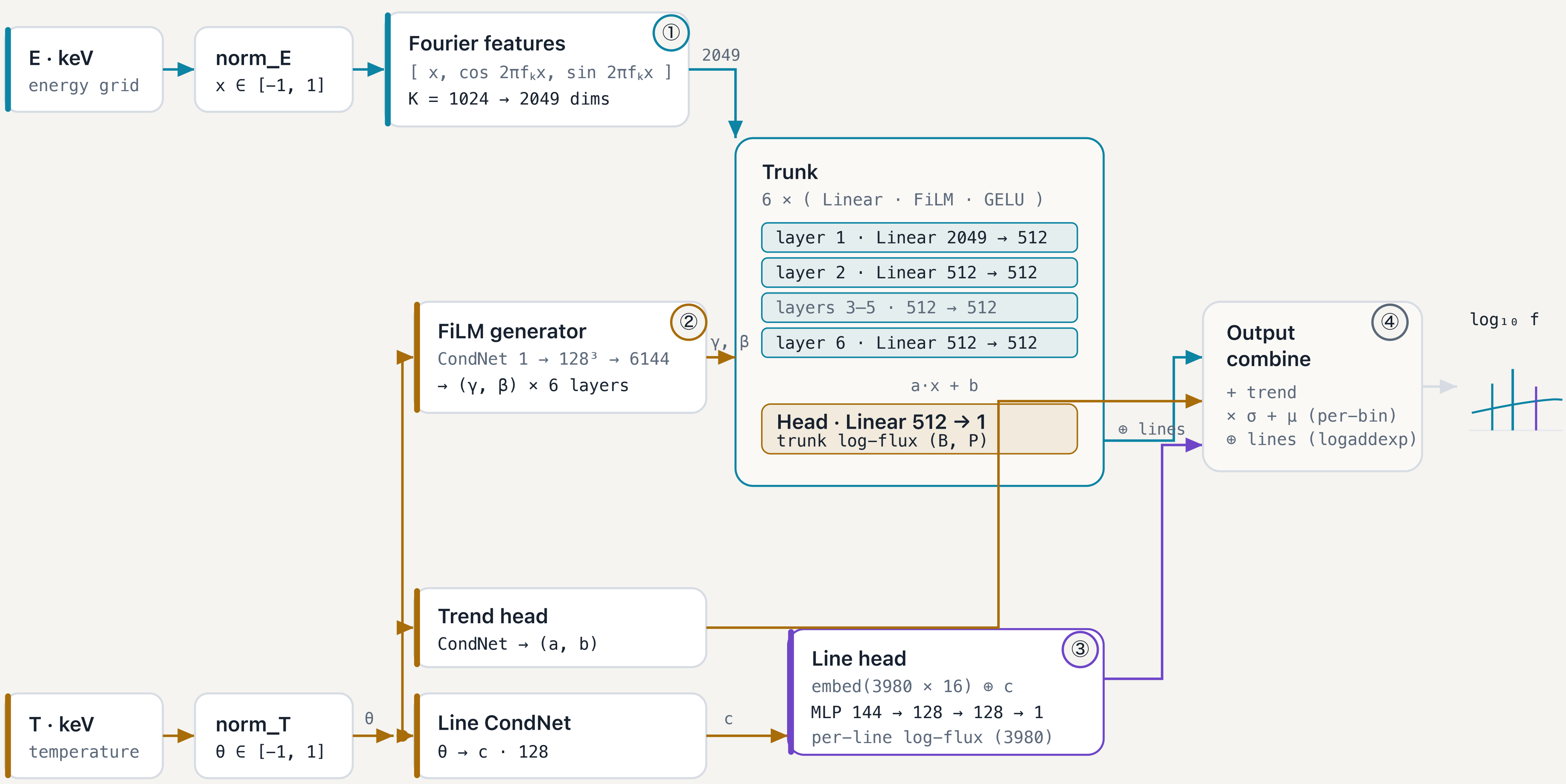
SPEXAI · ELEMENT-26 CIE EMULATOR · BIG_TRUNK CONFIG

`SpectralOperator` — network architecture

A coordinate-based operator that predicts log-flux density as a continuous function of energy, conditioned on plasma temperature. Two input streams stay separate — the **energy coordinate** builds the spectrum, the **temperature** shapes it — and meet only inside the trunk.

`params 3.32 M` `trunk 512 × 6` `Fourier K 1024 · f≤4000` `lines 3980` `activation GELU` `test MRE 0.13 %` `t / spectrum 21.5 ms`

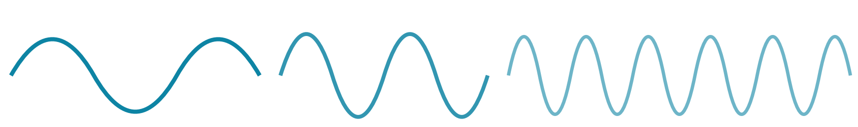
energy / coordinate path · temperature / conditioning path · line structure · ①–④ detail below



DETAIL ① · ENERGY ENCODING

Fourier feature embedding

A single scalar coordinate can't express one-bin emission lines. Each energy is lifted into 1024 sinusoids across four decades of frequency, so the trunk can resolve both the smooth continuum and sharp lines.

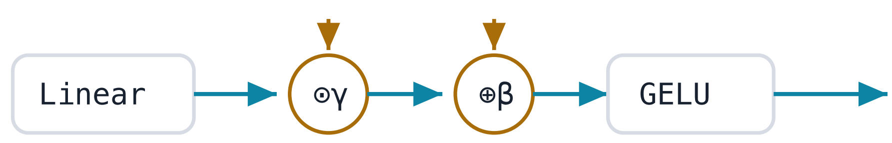


```
x ↦ [ x, cos 2πfₖx, sin 2πfₖx ]
fₖ log-spaced, f ∈ [0.25, 4000]
curriculum wₖ = clip(αK − k, 0, 1)
  → high freqs fade in over first 30%
```

DETAIL ② · CONDITIONING

FiLM-modulated trunk layer

Temperature never enters as an input feature. Instead it generates a scale and shift for every trunk layer, steering the shared spectral backbone toward the spectrum at that T.




```
h ← GELU( γ ⊙ Wh + β )
γ = 1 + gen(θ)   — identity at init
one (γ, β) pair per layer · 6 pairs total
```

DETAIL ③ · LINE STRUCTURE

Line-amplitude head

The 3980 known line energies are fixed. Each gets a learned embedding; an MLP turns [embedding, temperature code] into that line's flux, added to the continuum in *linear* space so lines never distort it.


```
out = logaddexp( trunk·ln10, line·ln10 ) / ln10
  = log₁₀( 10^trunk + 10^line )  — add fluxes
```

DETAIL ④ · RECONSTRUCTION

Output combination

Four terms assemble the final prediction per energy bin. Per-bin normalisation carries the static line forest in its mean, so the trunk only has to learn the temperature-dependent deviations.


```
f = ( head(h) + a·x + b )·σ_bin + μ_bin
then for line bins: ⊕ line head  (logaddexp)
```

Shapes: B = batch of spectra, P = sampled energy points (all 24 212 native bins at inference; a random subset while training). The trunk is evaluated once per energy and shared across all temperatures — conditioning enters only through the amber path. Velocity broadening and instrument response are applied *after* the network, analytically, and are not part of it. Configuration shown is `big_trunk` (Tier-2 winner); the deployed `t04` family uses a 384 × 5 trunk and K = 512.